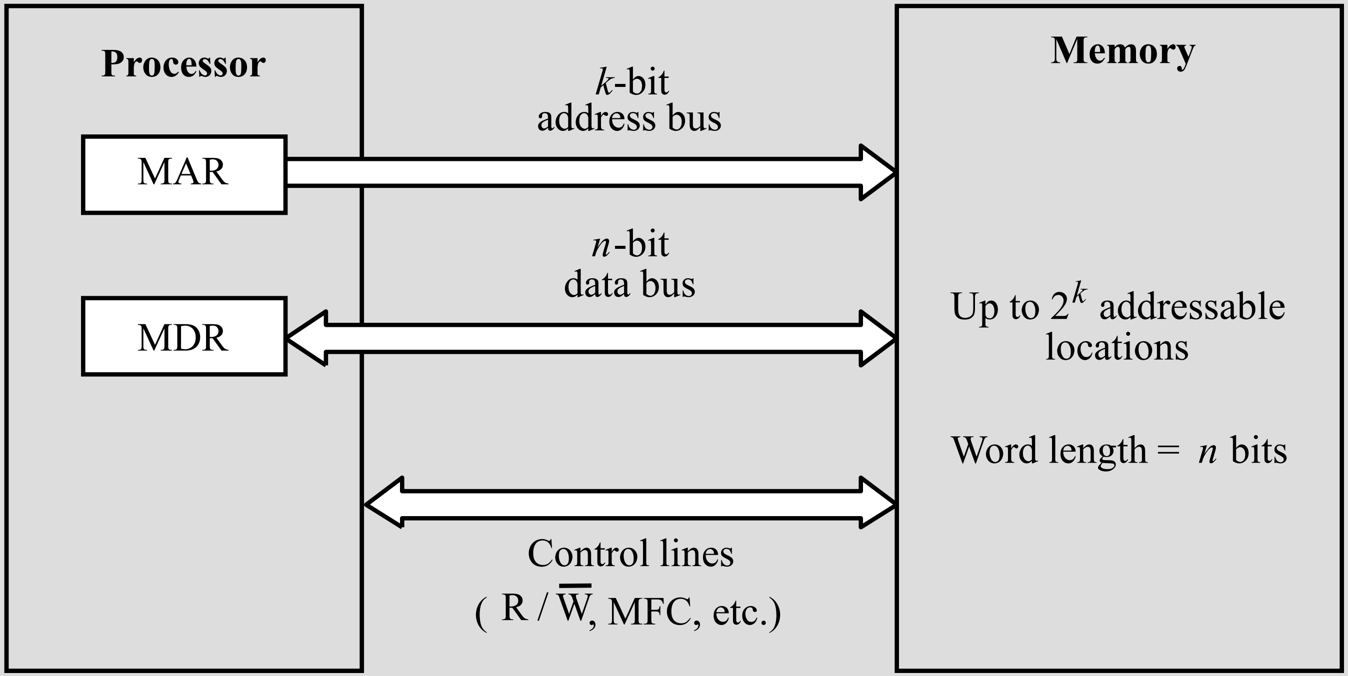
Microprocessor Systems and Interfacing

Topics covered:
Memory subsystem

Basic concepts

- Maximum size of the memory depends on the addressing scheme:
  - 16-bit computer generates 16-bit addresses and can address up to 2^{16} memory locations.
  - Number of locations represents the size of the address space of a computer.
- Most modern computers are byte-addressable.
- Memory is designed to store and retrieve data in word-length quantities.
  - Word length of a computer is commonly defined as the number of bits stored and retrieved in one memory operation.

Basic concepts (contd..)



Recall that the data transfers between a processor and memory involves two registers MAR and MDR.
If the address bus is k-bits, then the length of MAR is k bits.
If the word length is n-bits, then the length of MDR is n bits.
Control lines include R/$\overline{W}$ and MFC.
For Read operation R/$\overline{W}$ = 1 and for Write operation R/$\overline{W}$ = 0.

Basic concepts (contd..)

- Measures for the speed of a memory:
  - Elapsed time between the initiation of an operation and the completion of an operation is the memory access time (e.g. the time between the Read & MFC signals).
  - Minimum time between the initiation of two successive memory operations is memory cycle time (e.g. the time between two successive Read operations)
  - Memory Cycle time is slightly longer than memory access time
- In general, the faster a memory system, the costlier it is and the smaller it is.
- An important design issue is to provide a computer system with as large and fast a memory as possible, within a given cost target.
- Several techniques to increase the effective size and speed of the memory:
  - Cache memory (to increase the effective speed).
  - Virtual memory (to increase the effective size).

Semiconductor RAM memories

- Random Access Memory (RAM) memory unit is a unit where any location can be addressed in a fixed amount of time, independent of the location's address.
- Internal organization of memory chips:
  - Each memory cell can hold one bit of information.
  - Memory cells are organized in the form of an array.
  - One row is one memory word.
  - All cells of a row are connected to a common line, known as the "word line".
  - Word line is connected to the address decoder.
  - Sense/write circuits are connected to the data input/output lines of the memory chip.

Semiconductor RAM memories (contd..)

- Static RAMs (SRAMs):
  - Consist of circuits that are capable of retaining their state as long as the power is applied.
  - Volatile memories, because their contents are lost when power is interrupted.
  - Access times of static RAMs are in the range of few nanoseconds.
  - Capacity is low (each cell consists of 6 transistors)
  - However, the cost is usually high.
- Dynamic RAMs (DRAMs):
  - Do not retain their state indefinitely.
  - Contents must be periodically refreshed.
  - Contents may be refreshed while accessing them for reading.
  - Access time is longer than SRAM
  - Capacity is higher than SRAM (each cell consists of 1 transistor)
  - Cost is lower than SRAM

Semiconductor RAM memories (contd..)

- Data is transferred between the processor and the memory in units of single word or a block.
- Speed and efficiency of data transfer has a large impact on the performance of a computer.
- Two metrics of performance: Latency and Bandwidth.
- **Latency**: **Time taken to transfer a single word of data to or from memory**.
- **Bandwidth:** it is useful to define a performance measure in terms of **the number of bits or bytes transferred in one second.**
  - Bandwidth of a memory unit depends on:
    - **Speed** of access to the chip.
    - How many **bits** can be accessed in parallel.

Memory systems

- Various **factors** such as cost, speed, power consumption and size of the chip determine how a **RAM** is chosen for a given application.
- **Static RAMs**:
  - Chosen when **speed** is the primary concern.
  - **Circuit** implementing the basic cell is highly **complex**, so **cost** and **size** are affected.
  - Used mostly in **cache** memories.
- **Dynamic RAMs**:
  - Predominantly used for implementing computer **main** memories.
  - **High densities** available in these chips.
  - **Economically** viable for implementing large memories.

Read-Only Memories (ROMs)

- SRAM and SDRAM chips are volatile:
  - Lose the contents when the power is turned off.
- Many applications need memory devices to retain contents after the power is turned off.
  - For example, computer is turned on, the operating system must be loaded from the disk into the memory.
  - Store instructions which would load the OS from the disk.
  - Need to store these instructions so that they will not be lost after the power is turned off.
  - We need to store the instructions into a non-volatile memory.
- Non-volatile memory is read in the same manner as volatile memory.
  - Separate writing process is needed to place information in this memory.
  - Normal operation involves only reading of data, this type of memory is called Read-Only memory (**ROM**).

Read-Only Memories (contd..)

- Read-Only Memory:
  - Data are written into a ROM when it is manufactured.
- Programmable Read-Only Memory (PROM):
  - Allow the data to be loaded by a user.
  - Process of inserting the data is irreversible.
  - Storing information specific to a user in a ROM is expensive.
  - Providing programming capability to a user may be better.
- Erasable Programmable Read-Only Memory (EPROM):
  - Stored data to be erased and new data to be loaded.
  - Flexibility, useful during the development phase of digital systems.
  - Erasable, reprogrammable ROM.
  - Erasure requires exposing the ROM to UV light.

Read-Only Memories (contd..)

- Electrically Erasable Programmable Read-Only Memory (EEPROM):
  - To erase the contents of EPROMs, they have to be exposed to ultraviolet light.
  - Physically removed from the circuit.
  - EEPROMs the contents can be stored and erased **electrically**.

Flash memory

- Flash memory has similar approach to EEPROM.
- Read the contents of a single cell, but write the contents of an entire block of cells.
- Flash devices have greater density.
  - Higher capacity and low storage cost per bit.
- Power consumption of flash memory is very low, making it attractive for use in equipment that is battery-driven.
- Single flash chips are not sufficiently large, so larger memory modules are implemented using flash cards and flash drives.

Acronyms

RAM	--Random Access Memory	time taken to access any arbitrary location in memory is constant (c.f., disks)
ROM	--Read Only Memory	ROMs are RAMs which can only be written to once; thereafter they can only be read Older uses included storage of bootstrap info
PROM	--Programmable ROM	A ROM which can be bench programmed
EPROM	--Erasable PROM	A PROM which can be erased for rewriting
EEPROM	--Electrically EPROM	A PROM which can be erased electrically.
SRAM	--Static RAM	RAM chip which loses contents upon power off
DRAM	--Dynamic RAM	RAM chip whose contents need to be refreshed.
SDRAM	--Synchronous DRAM	RAM whose memory operations are synchronized with a clock signal.